ADS1298 #1

Electrodes

E1 R1 2KΩ R2 2KΩ C1 620pF C2 620pF AGND

E2 R3 2KΩ R4 2KΩ C3 620pF C4 620pF AGND

E3 R5 2KΩ R6 2KΩ C5 620pF C10 620pF AGND

E4 R7 2KΩ R8 2KΩ C11 620pF C12 620pF AGND

E5 R9 2KΩ R10 2KΩ C13 620pF C14 620pF AGND

E6 R11 2KΩ R12 2KΩ C15 620pF C16 620pF AGND

E7 R13 2KΩ R14 2KΩ C17 620pF C18 620pF AGND

E8 R15 2KΩ R16 2KΩ C21 620pF C22 620pF AGND

+3.3V +3.3V C6 0.1uF C7 10uF AGND

+3.3V +3.3V C8 0.1uF C9 10uF AGND

U1 ADS1298xPAG

IN1P IN1N IN2P IN2N IN3P IN3N IN4P IN4N IN5P IN5N IN6P IN6N IN7P IN7N IN8P IN8N

AVDD AVDD1 DVDD

WCT RLDOUT RLDIN RLDINV RLDREF +3.3V

TESTP_PACE_OUT1 TESTP_PACE_OUT2

GPIO1 GPIO2 GPIO3 GPIO4 AGND

DAISY_IN SCLK DIN CS DOUT

Unused, shorted AGND

SCLK MOSI CS_ADC1 MISO

CLK CLKSEL START DRDY PWDN RESET

U1 CLK_EN=1 required ADC_CLK +3.3V

DRDY_ADC1 PWDN RESET

RESV1 AGND

VCAP1 VCAP2 VCAP3 VCAP4

VCAP1_ADC1 VCAP2_ADC1 VCAP3_ADC1 VCAP4_ADC1

VREFP VREFN

VREFP_ADC1 C19 0.1uF C20 10uF AGND

AVSS AVSS1 AGND

C23 22uF C24 1uF C25 1uF C26 1uF AGND

ADS1298 #2

Electrodes

E9 R17 2KΩ R18 2KΩ C27 620pF C28 620pF AGND

E10 R19 2KΩ R20 2KΩ C29 620pF C30 620pF AGND

E11 R21 2KΩ R22 2KΩ C31 620pF C36 620pF AGND

E12 R23 2KΩ R24 2KΩ C37 620pF C38 620pF AGND

E13 R25 2KΩ R26 2KΩ C39 620pF C40 620pF AGND

E14 R27 2KΩ R28 2KΩ R29 2KΩ C41 620pF C42 620pF C43 620pF AGND

E15 R30 2KΩ R31 2KΩ C44 620pF C47 620pF AGND

E16 R32 2KΩ C48 620pF AGND

+3.3V +3.3V C32 0.1uF C33 10uF AGND

+3.3V +3.3V C34 0.1uF C35 10uF AGND

U2 ADS1298xPAG

IN1P IN1N IN2P IN2N IN3P IN3N IN4P IN4N IN5P IN5N IN6P IN6N IN7P IN7N IN8P IN8N

AVDD AVDD1 DVDD

WCT RLDOUT RLDIN RLDINV RLDREF +3.3V

TESTP_PACE_OUT1 TESTP_PACE_OUT2

GPIO1 GPIO2 GPIO3 GPIO4 AGND

DAISY_IN SCLK DIN CS DOUT

Unused, shorted AGND

SCLK MOSI CS_ADC2 MISO

CLK CLKSEL START DRDY PWDN RESET

U1 CLK_EN=1 required ADC_CLK

DRDY_ADC2 PWDN RESET

DRDY: duplicate only; MCU still uses U1 DRDY

RESV1 AGND

VCAP1 VCAP2 VCAP3 VCAP4

VCAP1_ADC2 VCAP2_ADC2 VCAP3_ADC2 VCAP4_ADC2

VREFP VREFN

VREFP_ADC2 C45 0.1uF C46 10uF AGND

AVSS AVSS1 DGND

C49 22uF C50 1uF C51 1uF C52 1uF AGND

+5V +3.3V +3.3V

U3 ESP32-S3-DevKitC

CHIP_PU 5V 3V3 GND

GPIO43/U0TXD GPIO44/U0RXD

GPIO15/ADC2_CH4/32K_P GPIO16/ADC2_CH5/32K_N

GPIO20/USB_D+ GPIO19/USB_D-

GPIO39/MTCK GPIO40/MTDO GPIO41/MTDI GPIO42/MTMS

GPIO0 GPIO1/ADC1_CH0 GPIO2/ADC1_CH1 GPIO3/ADC1_CH2 GPIO4/ADC1_CH3 GPIO5/ADC1_CH4 GPIO6/ADC1_CH5 GPIO7/ADC1_CH6 GPIO8/ADC1_CH7 GPIO9/ADC1_CH8 GPIO10/ADC1_CH9 GPIO11/ADC2_CH0 GPIO12/ADC2_CH1 GPIO13/ADC2_CH2 GPIO14/ADC2_CH3 GPIO17/ADC2_CH6 GPIO18/ADC2_CH7 GPIO21 GPIO35 GPIO36 GPIO37 GPIO38 GPIO45 GPIO46 GPIO47 GPIO48

RESET PWDN START DRDY_ADC1 CS_ADC1 MOSI SCLK MISO CS_ADC2

ESP32-S3-DevKitC AGND

F21: probe access only; test-pad footprints/layout pending

TP1 ADC_CLK ADC_CLK

TP2 DRDY_ADC1 DRDY_ADC1

TP3 DRDY_ADC2 DRDY_ADC2

TP4 3V3 +3.3V

TP5 BAT +3.7V

TP6 AGND AGND

TP7 VREFP_ADC1 VREFP_ADC1

TP8 VCAP1_ADC1 VCAP1_ADC1

TP9 VCAP2_ADC1 VCAP2_ADC1

TP10 VCAP3_ADC1 VCAP3_ADC1

TP11 VCAP4_ADC1 VCAP4_ADC1

TP12 VREFP_ADC2 VREFP_ADC2

TP13 VCAP1_ADC2 VCAP1_ADC2

TP14 VCAP2_ADC2 VCAP2_ADC2

TP15 VCAP3_ADC2 VCAP3_ADC2

TP16 VCAP4_ADC2 VCAP4_ADC2

Grounding Configuration

AGND AGND

ADC LDO Configuration

U4 LP2985-3.3

+3.7V C53 1uF EN VIN ON/OFF GND VOUT BP C55 10nF C54 2.2uF AGND

LP2985_OUT_DISABLED

F04: U4 output isolated with C54: no system load.
Both LDOs remain OFF. Qualify LP silicon/C54 before enable.

START R33 10k AGND

RESET R34 10k AGND

PWDN R35 10k AGND

F09: 10k pulldowns bias START, RESET and PWDN low while MCU is high-Z.
CS defaults / complete power sequencing remain unresolved.

V1.2 PARTIAL CORRECTIONS — NOT READY FOR POWER-UP OR WEARER USE
F04: U4/C54 isolated from main rail. F05: R38 holds both LDOs OFF; operating power unqualified.
F06/F18: restore U1 CLK_EN after RESET; allow U2 clock/POR; configure before shared START.
F09: START/RESET/PWDN plus DIN/SCLK have pulldowns; CS/full sequencing unresolved.
Enable/settle 2.4V reference; verify gain/rate; keep PACE/RLD/WCT and RLD rerouting off.
F07/F08: protected body bias and wearer protection remain unimplemented.

Battery Configuration

+3.7V BT1 Battery AGND

MCU LDO Configuration

U5 TLV75533PDBV

+3.7V C56 1uF EN IN OUT GND EN +3.3V C57 200uF AGND

MOSI R36 10k AGND

SCLK R37 10k AGND

EN R38 0

V1.2 F09: R36/R37 bias DIN and SCLK low; CS / full sequencing unresolved.
F05: R38 = OR FITTED: both LDOs held OFF. No functional enable source.
Keep R38 fitted until functional EN. F13/F15/F16/F17 qualification is complete.
USB / external +3.3V can still backfeed the rail; OFF strap is not isolation.

V1.2 F12/F14 — PROCUREMENT REQUIREMENTS (not qualified installed parts)
R1–R32: 2k unchanged; 0.1%, >=0.1W, derating/pulse review required.
All 32 input 620pF caps: C0G/NP0, 1%; electrode impedance/CMRR still unverified.
C25/C51 VCAP3: 1uF unchanged, >=10V; qualify capacitance at ~5.2V DC.
C55 BP: 10nF unchanged; leakage <100nA at operating bias/temperature.
No MPN, footprint, physical performance or wearer-safety qualification is implied.

Version 1.2
McGill AI Lab — Researcher: Karen Chen Lai
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